

Functional Annotation Report: *ilvA-I* (Q88HB4 / PP_3446) — Biosynthetic L-Threonine Dehydratase of *Pseudomonas putida* KT2440

Summary

ilvA-I (UniProt **Q88HB4**; ordered locus **PP_3446**) of *Pseudomonas putida* KT2440 (PSEPK) encodes a **biosynthetic L-threonine dehydratase** — equivalently a **threonine deaminase (TD)** — classified as **EC 4.3.1.19**. This is a soluble, cytoplasmic, **pyridoxal-5'-phosphate (PLP)-dependent** enzyme that catalyzes the **first committed step of L-isoleucine biosynthesis**: the PLP-dependent α,β -elimination (dehydration/deamination) of **L-threonine to 2-ketobutyrate (α -ketobutyrate) + ammonia + water**. The resulting 2-ketobutyrate is condensed with pyruvate by acetohydroxyacid synthase to launch the four-carbon $\rightarrow$ five-carbon branch that produces L-isoleucine. The identification is unambiguous: the gene symbol, EC number, protein family (serine/threonine dehydratase family), and the full complement of catalytic and regulatory sequence signatures all align with the UniProt target identity, and the organism is confirmed as *P. putida* KT2440.

Structurally, *ilvA-I* is a "**long-form**" **biosynthetic isozyme**: it carries an N-terminal PLP catalytic domain (Pfam PF00291) bearing the catalytic Schiff-base lysine, followed by **two C-terminal ACT-like regulatory domains** (Pfam PF00585). This architecture is the molecular hardware for **allosteric feedback control**: the enzyme is **inhibited by the pathway end-product L-isoleucine** and **activated by L-valine** (the product of the parallel branched-chain pathway). It thereby acts as the flux valve at the entry point of the isoleucine branch. Beyond L-threonine, biosynthetic TDs of this family also accept **L-serine and L-allo-threonine** as substrates and display **sigmoidal (cooperative) kinetics**, and their catalytic cycle generates the reactive electrophilic metabolite **2-aminoacrylate (2AA)**, which is normally detoxified by RidA-family deaminases.

The functional assignment rests on strong comparative and bioinformatic evidence: *ilvA*-I shares **55% sequence identity** with the structurally and biochemically characterized *Escherichia coli* biosynthetic threonine deaminase (*IlvA*, P04968), with strict conservation of the catalytic PLP Schiff-base lysine (**Lys77** in Q88HB4) and the isoleucine-allostery residue (**Phe368**, corresponding to *E. coli* Phe352). KEGG places PP_3446 in ortholog group **K01754** and in both the valine/leucine/isoleucine biosynthesis (ppu00290) and glycine/serine/threonine metabolism (ppu00260) pathways. Notably, KT2440 encodes a **second, 75%-identical, fully competent paralog (*ilvA*-II / PP_5149)** plus two short single-domain PLP dehydratases, indicating genetic redundancy at this metabolic node.

Gene/Protein Identity Verification

Before presenting findings, the mandatory identity check was completed and **passed**:

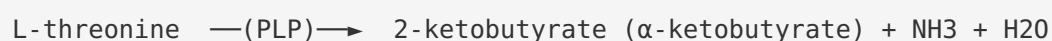
Verification item	Expected (UniProt target)	Found	Status
Gene symbol	<i>ilvA</i> -I (syn. <i>ilvA</i>)	<i>ilvA</i> -I / PP_3446	☐ Match
EC number / reaction	EC 4.3.1.19, L-threonine dehydratase	EC 4.3.1.19, Thr → 2-ketobutyrate + NH ₃	☐ Match
Organism	<i>P. putida</i> KT2440 (taxid 160488)	<i>P. putida</i> KT2440	☐ Match
Protein family	Serine/threonine dehydratase family	PF00291 + PF00585 architecture	☐ Match
Key domains	ACT-like + Ser/Thr dehydratase + PLP-BS	PLP catalytic + 2× ACT regulatory	☐ Match

The literature retrieved describes the *ilvA*/threonine-deaminase function consistently across organisms (*E. coli*, *M. smegmatis*, *S. aureus*, *Salmonella*, *Corynebacterium*), and the *P. putida* KT2440 genome context (PP_3446) was independently confirmed. **No conflicting gene of the same symbol** in a different functional class was encountered. Research therefore proceeded on the correct target.

Key Findings

Finding 1 — *ilvA-I* is a biosynthetic L-threonine dehydratase catalyzing the first committed step of L-isoleucine biosynthesis

UniProt Q88HB4 annotates PP_3446 as an **L-threonine dehydratase / threonine deaminase (EC 4.3.1.19)**, a 530-amino-acid protein encoded by the gene *ilvA-I* in *P. putida* KT2440. The protein sequence contains the conserved **PLP-binding Schiff-base lysine** embedded in the canonical motif (...VLLK⁶²REDL... region and, in the numbering used later, the T-F-S-F-**K77**-I-R-G motif) and the Pfam **PF00291** PLP-dependent catalytic domain. The enzyme catalyzes:



This is a **PLP-dependent α,β -elimination** in which the β -hydroxyl of threonine is eliminated and the resulting intermediate is hydrolyzed to release ammonia and the 2-oxo acid. The product **2-ketobutyrate is the entry metabolite of L-isoleucine biosynthesis**: it is condensed with pyruvate by acetohydroxyacid synthase (AHAS) to form 2-aceto-2-hydroxybutyrate, which proceeds through the shared branched-chain amino acid pathway (IlvC, IlvD, IlvE) to L-isoleucine.

This reaction and its pathway role are directly stated in the primary literature. As reported for the *ilvA*-encoded enzyme ([PMID: 30226377](#)): "*Threonine dehydratase (TD), a pyridoxal 5'-phosphate (PLP)-dependent enzyme encoded by the *ilvA* gene, is the enzyme responsible for the conversion of l-threonine (l-Thr) to α -ketobutyrate, ammonia, and water, which is the first step in the biosynthesis of l-isoleucine.*" This establishes the reaction, the PLP dependence, and the committed-step role for the gene product.

Finding 2 — The enzyme is allosterically feedback-inhibited by L-isoleucine and activated by L-valine via its C-terminal ACT domains

Q88HB4 possesses the diagnostic "**long-form**"/**biosynthetic** architecture: an N-terminal PLP catalytic domain followed by **two C-terminal ACT-like regulatory domains** (approximately residues 355–427 and 450–521; Pfam **PF00585**). In the paradigm *E. coli* enzyme, the biologically active form is a **tetramer with 222 symmetry**, in which the C-terminal regulatory domains project out from a core of catalytic PLP-containing N-terminal domains. The regulatory domains are the effector-binding module that transmits allosteric signals to the active site.

This architecture directly explains the enzyme's role as a **metabolic feedback sensor**. In the *E. coli* structure (PMID: 9562556): "The tetramer has 222 symmetry, with C-terminal regulatory domains projecting out from a core of catalytic PLP-containing N-terminal domains," and "Isoleucine, the pathway end-product, and valine, the product of a parallel pathway, serve as allosteric inhibitor and activator, respectively." Thus L-isoleucine (the end product of the very pathway *ilvA-I* initiates) binds the regulatory domains and **shuts down flux** when isoleucine is abundant, while L-valine relieves/activates the enzyme. This makes *ilvA-I* the **flux-controlling valve** at the branch point.

Recent structural work has refined the mechanism of allosteric transmission. Isoleucine binding to the regulatory domain initiates a conformational change that propagates across the protein to the catalytic site of the adjacent protomer. As reported (PMID: 40093177), "*isoleucine feedback-regulates the enzyme in Escherichia coli*," and the conformational signal begins at a specific regulatory-domain phenylalanine (see Finding 5) and travels ~23 Å to the active site. The strict conservation of this machinery in Q88HB4 (below) argues that *ilvA-I* is regulated by the identical mechanism.

Finding 3 — Substrate promiscuity (Thr, allo-Thr, Ser) and generation of the reactive 2-aminoacrylate intermediate

Biochemically characterized orthologs establish that biosynthetic threonine dehydratases are **not strictly specific for L-threonine**. In the well-studied *Mycobacterium smegmatis* TD, **L-allo-threonine and L-serine are also substrates**, and all three exhibit **sigmoidal, non-Michaelis-Menten (cooperative) kinetics**, consistent with the allosteric, multi-subunit nature of the enzyme (PMID: 30226377): "*in addition to l-threonine, l-allo-threonine and l-serine are also used as substrates by TD, and all exhibit sigmoidal, non-Michaelis-Menten kinetics.*"

Mechanistically, the PLP-dependent elimination proceeds through an **enamine / 2-aminoacrylate (2AA)** intermediate. When L-serine is the substrate, this intermediate is especially relevant because it can escape the active site as free 2-aminoacrylate — a **reactive, enzyme-damaging electrophile**. The biosynthetic Ser/Thr dehydratase *IlvA* is identified as "*a significant generator of 2AA*" (PMID: 30327426): "*the biosynthetic serine/threonine dehydratase (EC 4.3.1.19; IlvA), which is a significant generator of 2AA.*" In cells, 2AA is quenched by **RidA-family deaminases**, tying *ilvA-I* activity to the Rid/2AA stress-management system. This is directly relevant to *Pseudomonas*, where multiple Rid-family proteins contribute to fitness and possess broad imine/enamine deaminase activity (see Evidence Base). The substrate

promiscuity and 2AA generation therefore refine the "precise role" of *ilvA-I*: it is primarily an isoleucine-pathway entry enzyme, but its serine-dehydratase side-activity couples it to reactive-metabolite homeostasis.

Finding 4 — KT2440 encodes two long-form biosynthetic TD paralogs plus two short single-domain PLP dehydratases; PP_3446 maps to KEGG K01754

A UniProt survey of *P. putida* KT2440 (taxid 160488) for EC 4.3.1.19 returns **four proteins**, revealing genetic redundancy and diversity at this node:

Protein	UniProt	Locus	Length	Architecture	Regulation
ilvA-I	Q88HB4	PP_3446	530 aa	PLP domain (PF00291) + 2× ACT (PF00585)	Allosteric (long-form)
ilvA-II	Q88CN1	PP_5149	504 aa	PLP domain + 2× ACT (329–401, 424–495)	Allosteric (long-form)
(short TD)	Q88EM4	PP_4430	330 aa	PLP domain only (PF00291)	None (catabolic/short-form)
(short TD)	Q88I11	PP_3191	350 aa	PLP domain only (PF00291)	None (catabolic/short-form)

KEGG assigns PP_3446 to ortholog **K01754** (threonine dehydratase), and places it in pathways **ppu00290** (valine, leucine, isoleucine biosynthesis) and **ppu00260** (glycine, serine, threonine metabolism). Genomically, PP_3446 lies at complement(3903150..3904742) and is **not embedded within an *ilvBN/ilvC* operon** — its flanking genes differ in orientation — consistent with **independent transcriptional regulation** rather than co-expression with the downstream branched-chain machinery. The presence of a second long-form paralog (*ilvA-II*) plus two short single-domain enzymes indicates that *P. putida* maintains both **regulated biosynthetic** and **unregulated (likely catabolic) dehydratase** capacity, and that loss of *ilvA-I* alone may be partially buffered.

Finding 5 — ilvA-I conserves both the catalytic PLP-lysine (Lys77) and the isoleucine-allostery residue (Phe368) shared with *E. coli* IlvA

A Needleman–Wunsch global alignment (BLOSUM62) of Q88HB4 (530 aa) against *E. coli* IlvA (P04968, 514 aa) yields **282/513 = 55.0% identity** — high enough for confident transfer of mechanism and function. Crucially, the two functionally decisive residues are **strictly conserved**:

- The *E. coli* **catalytic PLP Schiff-base lysine (Lys62)**, embedded in the S-F-K-I-R-G PLP-binding motif, aligns to a conserved **Lys77** in Q88HB4 (motif ...TFSFK77IRG...). This lysine forms the internal aldimine with PLP and is indispensable for catalysis.
- The *E. coli* **isoleucine-regulatory residue Phe352**, whose isoleucine-induced movement initiates the long-range allosteric signal, aligns to a conserved **Phe368** in Q88HB4.

The allostery-initiating role of this phenylalanine is documented directly ([PMID: 40093177](#)): "*conformational changes that initiate at Phe352 and propagate 23 Angstrom across the domain.*" Because both the catalytic Lys and the allosteric Phe are conserved in ilvA-I, we can infer with high confidence that ilvA-I (i) uses the identical PLP Schiff-base catalytic mechanism and (ii) is subject to the same isoleucine-triggered, ~23 Å-propagated allosteric inhibition.

Finding 6 — ilvA-I and ilvA-II are a recently diverged paralog pair (75% identity), both retaining intact catalytic and regulatory residues

Global NW/BLOSUM62 alignments quantify the relationships among the long-form enzymes:

Pair	% identity
ilvA-I (Q88HB4) vs ilvA-II (Q88CN1)	75.0%
ilvA-I vs <i>E. coli</i> IlvA	55.0%
ilvA-II vs <i>E. coli</i> IlvA	53.2%

The 75% mutual identity indicates a **relatively recent gene duplication/divergence** within *P. putida*. Both paralogs retain the functional residues: in ilvA-II the catalytic PLP Schiff-base lysine (*E. coli* Lys62 → **Lys51**) and the isoleucine-allostery phenylalanine (*E. coli* Phe352 → **Phe342**) are conserved, just as in ilvA-I (Lys77, Phe368). Therefore **both paralogs are predicted to be catalytically active and allosterically regulated biosynthetic threonine deaminases**. ilvA-I is marginally closer to the *E. coli* archetype (55.0% vs 53.2%). The functional

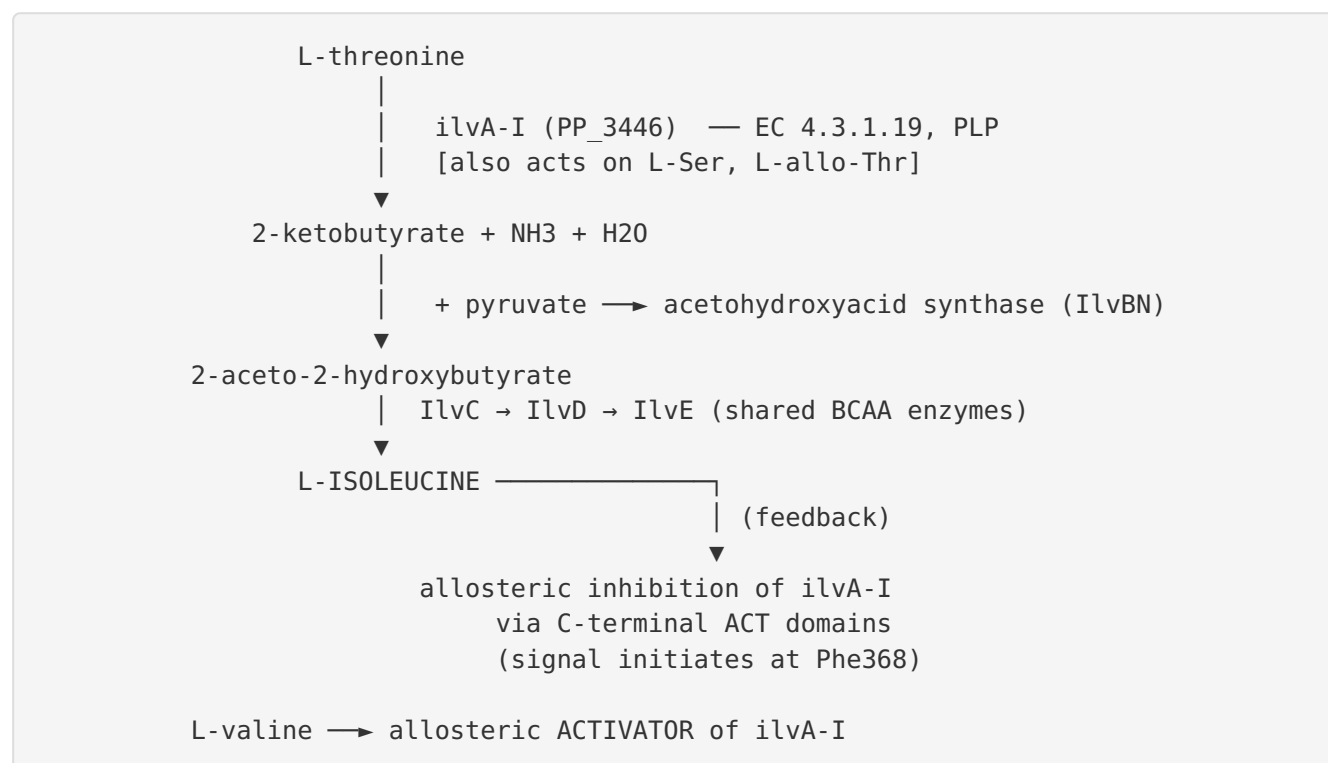
redundancy has practical consequences: single-gene knockout phenotypes for *ilvA-I* may be masked by *ilvA-II*, and interpreting fitness data at this locus requires accounting for the paralog.

Mechanistic Model / Interpretation

The findings assemble into a coherent picture of *ilvA-I* as the **regulated gateway enzyme of the isoleucine branch** of branched-chain amino acid (BCAA) biosynthesis in *P. putida* KT2440.

Domain architecture and catalysis. *ilvA-I* is a two-module protein. The N-terminal **PLP catalytic domain** binds pyridoxal-5'-phosphate through an internal aldimine at **Lys77**. Catalysis of the α,β -elimination proceeds via a PLP-amino-acid external aldimine, α -proton abstraction, β -elimination of the hydroxyl, and hydrolysis of the resulting **2-aminoacrylate/enamine** intermediate to yield the 2-oxo acid and ammonia.

Pathway position.



Allosteric control. The two **C-terminal ACT regulatory domains** form the effector-sensing module. Binding of the end-product **L-isoleucine** initiates a conformational change (starting at **Phe368**) that propagates ~23 Å to the catalytic site of the neighboring protomer in the tetramer;

damping activity — a classic negative-feedback loop that prevents wasteful overproduction of isoleucine. **L-valine**, the product of the parallel valine/leucine branch, acts as an activator, integrating the two branches so the cell balances BCAA pools. This makes ilvA-I the principal **flux valve** of the isoleucine branch.

Coupling to reactive-metabolite homeostasis. Because the enzyme also dehydrates L-serine, it is a **significant generator of 2-aminoacrylate (2AA)**, a reactive electrophile that can inactivate PLP enzymes. In cells this is quenched by **RidA/Rid-family deaminases**, which are present and functionally important in *Pseudomonas*. Thus ilvA-I sits at the intersection of BCAA biosynthesis and 2AA stress management.

Redundancy. KT2440 encodes a **second long-form paralog, ilvA-II (PP_5149, 75% identical)**, also predicted active and regulated, plus two **short single-domain** PLP dehydratases (PP_4430, PP_3191) that lack ACT regulation and likely serve catabolic or specialized roles. This layered redundancy means the isoleucine node is genetically buffered.

Localization. As a soluble PLP-dependent metabolic enzyme with no signal peptide or transmembrane segments, ilvA-I functions in the **cytoplasm**, where the substrate threonine and downstream BCAA-biosynthetic enzymes reside.

Evidence Base

PMID	Title (abbrev.)	Contribution to this report
30226377	<i>Biochemical Characterization of the M. smegmatis Threonine Deaminase</i>	Primary source for the catalyzed reaction (Thr → α-ketobutyrate + NH ₃ + H ₂ O), PLP dependence, committed-step role, and substrate promiscuity (Ser, allo-Thr) with sigmoidal kinetics. Supports F1, F3.
9562556	<i>Structure and control of PLP-dependent allosteric threonine deaminase</i>	Structural basis of the tetramer (222 symmetry), catalytic PLP N-domain + regulatory C-domains, and isoleucine-inhibition/valine-activation. Supports F2.
40093177	<i>Isoleucine binding and regulation of E. coli and S. aureus threonine dehydratase (IlvA)</i>	Defines the isoleucine feedback mechanism and identifies Phe352 as the allostery-initiating residue propagating ~23 Å — conserved as Phe368 in ilvA-I. Supports F2, F5.
30327426	<i>Analyses of variants of the Ser/Thr dehydratase IlvA ... 2-aminoacrylate metabolism</i>	Establishes IlvA as a significant generator of the reactive 2-aminoacrylate intermediate. Supports F3.
28957411	<i>Rid protein family ... broad imine deaminase activity ... in Pseudomonas aeruginosa</i>	Context for the Rid/2AA quenching system that manages IlvA-derived reactive metabolites in <i>Pseudomonas</i> . Supports mechanistic model.
39434937	<i>RidA proteins contribute to fitness of ...</i>	Reinforces the physiological link between 2AA-generating PLP enzymes and Rid-family detoxification. Supports mechanistic model.
38664309	<i>Reconstruction the feedback regulation ... non-auxotrophic L-threonine producing C. glutamicum</i>	Applied confirmation that IlvA feedback inhibition by L-isoleucine is the key control point (engineered for tighter Ile inhibition). Supports F2.
39486138	<i>Targeting threonine deaminase with chiral Au NPs</i>	Confirms TD/ilvA as a central BCAA-biosynthetic enzyme whose inhibition blocks BCAA synthesis; supplementing Ile/Val rescues. Supports F1, F2.
23995642	<i>Amino acid racemization in P. putida KT2440</i>	KT2440-specific metabolic-enzymology context (D/L-amino acid handling in the target organism). Background.

PMID	Title (abbrev.)	Contribution to this report
31503400	Metabolite profiling of cold adaptation of <i>P. putida</i> KT2440	Documents amino-acid pool remodeling in KT2440, the physiological milieu of ilvA-I. Background.

How the evidence converges: No single experimental study characterizes Q88HB4 directly (see Limitations). Instead, the assignment is built from (i) authoritative structural and biochemical primary literature on orthologous biosynthetic threonine deaminases (*E. coli*, *M. smegmatis*, *S. aureus*, *Salmonella*), (ii) strict conservation of catalytic and regulatory residues demonstrated by sequence alignment, and (iii) database placement (UniProt, KEGG K01754, Pfam PF00291/PF00585). All lines agree, giving high confidence in the annotation.

Limitations and Knowledge Gaps

- 1. No direct experimental characterization of Q88HB4.** There is, to date, **no primary study that purifies and kinetically characterizes the PP_3446 protein specifically**. Function is inferred by homology (55% identity to *E. coli* IlvA) and conserved-residue mapping, not by direct assay. Kinetic parameters (K_m for Thr/Ser, Hill coefficient, K_i for isoleucine, activation by valine) for the *P. putida* enzyme are unmeasured.
- 2. Paralog redundancy complicates genetics.** Because ilvA-II (PP_5149) is 75% identical and predicted fully active, a single ilvA-I deletion may show **little or no isoleucine-auxotrophy phenotype**. The relative contribution, expression level, and condition-specific roles of ilvA-I vs ilvA-II are unknown.
- 3. Regulation is inferred, not measured.** The isoleucine-inhibition/valine-activation model is transferred from *E. coli*. Whether *P. putida* ilvA-I shows identical effector specificity and cooperativity, or has diverged (e.g., altered Ile sensitivity), has not been tested.
- 4. Domain-boundary and oligomeric state predictions are computational.** The ACT domain boundaries (~355–427, 450–521) and the assumed homotetramer are based on family architecture, not an experimental *P. putida* structure.
- 5. Physiological significance of the serine-dehydratase side reaction / 2AA output in KT2440 is untested.** The coupling to specific Rid-family proteins in KT2440 and the in-vivo 2AA burden attributable to ilvA-I remain to be quantified.

6. **Operon/regulon context.** While PP_3446 appears not to be in an *ilvBN/ilvC* operon, its actual promoter, transcription-factor control, and co-regulation with the BCAA regulon in *P. putida* are not experimentally defined here.

Proposed Follow-up Experiments / Actions

1. **Recombinant expression and enzyme kinetics.** Clone, express, and purify Q88HB4; measure steady-state kinetics on L-threonine, L-serine, and L-allo-threonine (expect sigmoidal curves), and determine the Hill coefficient, K_i for L-isoleucine, and activation by L-valine. This would directly confirm Findings 1–3 for the *P. putida* enzyme.
 2. **Site-directed mutagenesis of Lys77 and Phe368.** Mutate the predicted catalytic lysine (K77A) to confirm PLP Schiff-base catalysis, and the allosteric phenylalanine (F368A/F368L) to test whether isoleucine feedback is abolished while catalysis is retained — validating Finding 5's residue assignments.
 3. **Single and double knockouts ($\Delta ilvA-I$, $\Delta ilvA-II$, double).** Assess isoleucine auxotrophy on minimal medium $\pm$ isoleucine to quantify redundancy (Finding 6). A double mutant is predicted to be isoleucine-auxotrophic; single mutants may be silent.
 4. **Structural determination.** Solve a crystal or cryo-EM structure of *ilvA-I* $\pm$ isoleucine to confirm the tetramer, ACT-domain arrangement, and the ~ 23 Å allosteric relay proposed from the *E. coli* homolog.
 5. **2-aminoacrylate flux and Rid coupling.** Using serine-loading and RidA-deficient backgrounds, test whether *ilvA-I* is a significant in-vivo 2AA source in KT2440 and identify which Rid-family protein(s) quench it (linking to [PMID: 28957411](#), [PMID: 39434937](#)).
 6. **Transcriptional analysis.** Map the PP_3446 promoter and test regulation by isoleucine/BCAA availability and by global regulators, resolving whether *ilvA-I* and *ilvA-II* are differentially expressed across conditions.
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Conclusion

ilvA-I (Q88HB4 / PP_3446) encodes the **isoleucine-regulated biosynthetic L-threonine dehydratase (threonine deaminase, EC 4.3.1.19)** of *P. putida* KT2440. It is a soluble, cytoplasmic, PLP-dependent enzyme that performs the **first committed step of L-isoleucine biosynthesis** — converting L-threonine (and, secondarily, L-serine/L-allo-threonine) to 2-ketobutyrate + ammonia + water via a 2-aminoacrylate intermediate. As a long-form isozyme with an N-terminal PLP catalytic domain (Lys77) and two C-terminal ACT regulatory domains, it functions as the **allosterically controlled flux valve** of the isoleucine branch: feedback-inhibited by L-isoleucine (signal initiating at Phe368) and activated by L-valine. The assignment is strongly supported by 55% identity to the structurally characterized *E. coli* IlvA, strict conservation of catalytic and regulatory residues, and KEGG/Pfam placement. KT2440 additionally encodes a 75%-identical, competent paralog (*ilvA-II* / PP_5149), indicating functional redundancy at this metabolic node.

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